



**MIRAMAR LABS**

HYBRID ON-PREM / GCP AI PLATFORM

# Alpaca Options Trading Agents

A three-agent trading floor that turns a directional call into a specific options contract via an Alpaca MCP tool-calling agent — every LLM decision running as local inference on one NVIDIA DGX Spark — a machine running on 100% solar power.

Alpaca AI Trading Agents Hackathon · `lablab.ai` · Sep 2026 · paper account PA3MAA2HJLUG · \$100,000



POWERED BY

**NVIDIA DGX Spark**

ON 100% SOLAR POWER

# A directional signal is **not a trade**

"AAPL looks like a buy" is the easy half. Turning it into an options position is a second, harder decision — and normally a human options trader's job.

- **Which expiration?** Too short and theta eats you; too long and you overpay for time you don't need.
- **Which strike?** The delta has to match the conviction — a 0.30 lotto ticket and a 0.55 directional play are different trades.
- **Is the premium sane?** Wide spreads, thin open interest, and IV crush around events turn a good call into a losing fill.
- **Does it fit the book?** Notional exposure, existing positions, macro-event timing — all before a single order.

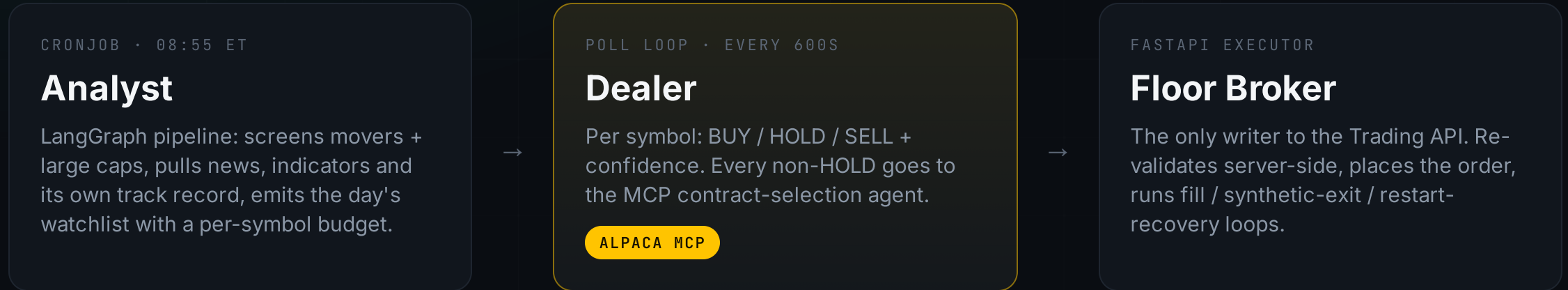
# Put an **LLM agent** in the contract-selection seat

Give it live read access to Alpaca's option chain, quotes and Greeks through MCP and let it reason to a specific contract — with a deterministic layer underneath that assumes the model can be **wrong or adversarial**.

- **The agent** browses the chain and proposes one contract: strike, expiration, delta, premium, and its reasoning.
- **The code** re-derives every hard constraint from the chain data itself — the model's numbers are never trusted, only its rationale.
- **A structured-output miss** degrades to a deterministic picker, never to “no trade.”

```
signal → MCP agent
      (chain / quotes / greeks)
      |
      ▼
      contract pick
      |
      validate → reconcile → re-validate
      |
      ▼
      order
```

# Three agents, one continuous floor



- State passes through a **portfolio ConfigMap** (Analyst → Dealer) and **Postgres** — the agents' memory across cycles: the Analyst and Dealer read their own past picks, decisions and execution outcomes back into their prompts. Also the audit trail and order reconciliation.
- Runs live on `k3s`, namespace `alpaca-options-trader` — not a backtest, not a one-off demo script.

# The **Alpaca MCP** integration

- `mcp_options.py` launches `alpaca-mcp-server` over stdio via `langchain-mcp-adapters`.
- **Read-only toolset** — `assets`, `options-data`, `account`. The Dealer can inspect the chain but **can never place an order**.
- Inputs: the underlying, the direction (call = bullish, put = bearish; always long), and the config DTE / delta windows.
- The LLM drives a **bounded  $\leq 6$ -round** tool-calling loop over live chain / quotes / Greeks.
- Returns a structured pick: strike, expiration, delta, mid-price premium, reasoning.

```
round 1  get_option_chain(AAPL)
round 2  filter DTE 15-45
round 3  get_quotes(candidates)
round 4  get_greeks(shortlist)
round 5  compare delta / spread
round 6  → OptionContractPick
        { strike, exp, delta,
          premium, reasoning }
```

# Validate → reconcile → re-validate

Everything downstream of the model call assumes the model is adversarial, not just wrong.

- 1 **Validate** — reject a pick that claims the wrong option type for the signal, or names a contract that was never seen in a chain response (a hallucination).
- 2 **Reconcile** — overwrite strike / expiration / delta / premium with what the chain actually showed for that exact contract. Only the reasoning text is trusted from the model.
- 3 **Re-validate** — server-side in Floor Broker, against the live DTE / delta windows and a fresh ask quote, immediately before the order is submitted.

This also blocks the interesting failure modes for free: a real contract with a **fabricated premium**, or a call-labelled pick whose OCC symbol is actually a put.

# Eight gates between “BUY” and an order

Each one deterministic — none of them is an LLM call.

- 1 Confidence threshold on the signal
- 2 Macro-event blackout calendar (FOMC / CPI / jobs / PCE)
- 3 Same-symbol stop-loss cooldown
- 4 Portfolio win-rate throttle
- 5 Daily profit-target / loss-limit halt
- 6 Operator kill switch — flipped in config, no redeploy
- 7 DTE / delta window re-validation
- 8 Hard notional cap, priced against the live market

Once a position is open, Alpaca has no server-side bracket for options — so Floor Broker runs a **30-second poll loop** that force-closes on a **50% loss**, a **175% gain**, or expiration reaching **3 DTE**, whichever comes first, regardless of whether new entries are currently gated.

# Everything on **one NVIDIA DGX Spark**

## k3s cluster

All four workloads in namespace `alpaca-options-trader` — Analyst, Dealer, Floor Broker, EOD report.

## Ollama `qwen2.5:32b-instruct`

Local inference behind every Analyst / Dealer / MCP decision — on a DGX that runs on 100% solar power. No external LLM API.

## Postgres

Shared platform instance — the agents' memory across cycles (past picks, decisions and outcomes feed the next run), plus audit trail and reconciliation.

## GitHub Actions runner

Self-hosted. push → test + lint → build 4 images → rolling deploy.

`config.yaml` is fetched fresh from GitHub every 60 seconds by every pod — risk thresholds, blackout dates, even the LLM model are a plain `git push`, no rebuild.

# Live, and behaving

**7**OPTION POSITIONS  
100% FILLED < 30s**\$100,319**EQUITY AT CLOSE  
+0.3% VS \$100k OPEN**+\$1,678**DAY-2 SESSION P/L  
+1.70% ON THE DAY**4 / 7**POSITIONS GREEN  
BEST V 375C +25%

- Two sessions of 600s polls — mostly HOLD, ~two dozen BUY signals produced 7 positions; the rest were **stopped by the risk layer** — seven sub-0.6 confidence, several duplicates, three with no chain contract passing the DTE / delta / OI / volume gates.
- **Gates observed firing:** the daily profit-target halt tripped repeatedly once intraday P/L cleared +\$1,000; confidence, duplicate and contract-quality gates each rejected a real signal.
- No synthetic exit has triggered — no position at -50%, +175%, or 3 DTE. Nothing closed yet, so no realized P/L or win rate. Live P/L + model badges in the README.

# Structured output is the **hard part** — not the MCP plumbing

- The tool-calling loop works every cycle: the agent calls the chain / quote / Greeks tools and pulls real data.
- But the closing `.with_structured_output()` returned **empty every cycle** on the first model, `qwen3.6:35b-a3b` — a 35B MoE with only ~3B active. Root cause was that active-parameter count, not context: the failing prompt was ~4k tokens against a 32k window.
- All 7 day-1-2 contracts came from the fallback path. **Fixed inside the window:** each candidate run through the real selection path against live MCP data (committed as `scripts/check_contract_selection.py`) — switched to dense `qwen2.5:32b-instruct-q4`, which returns valid structured picks (`fallback=False` on NVDA / TSLA / AAPL). Selection now runs on the model.

**`_fallback_pick`** stays a first-class safety net: it selects from the exact chain rows the loop already fetched, under the same gates — any future empty result degrades to “the code skims the chain like a human would,” never to nothing. The bulletproof long-term answer is a guided-decoding backend (vLLM / SGLang / NIM) — an infra migration off Ollama that didn't fit the window.

# An **LLM in the loop**, a deterministic layer under it

REPO [github.com/miramar-labs/alpaca-options-trading-agents](https://github.com/miramar-labs/alpaca-options-trading-agents)

LICENSE MIT, public — real, dated commits across the competition week

ACCOUNT Dedicated Alpaca paper account [PA3MAA2HJLUG](#) — \$100,000, trading continuously since 1 Sep

STACK Alpaca · Model Context Protocol · alpaca-mcp-server · LangGraph · Ollama · FastAPI · PostgreSQL · k3s

